MIL-R-3471C
29 June 1978
SUPERSEDING
MIL-R-3471B
20 July 1970

MILITARY SPECIFICATION

RULERS, DRAFTING, ADJUSTABLE CURVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adjustable curve rulers.

1.2 Classification. Rulers shall be of the following types and lengths, as specified (see 6.2):

Type I - Rubber edge.
  Length - 15 inches.
  Length - 31 inches.
Type II - Metal edge.
  Length - 15 inches.
  Length - 20 inches.
  Length - 25 inches.
  Length - 30 inches.
Type III - Plastic edge.
  Length - 12 inches.
  Length - 18 inches.
  Length - 24 inches.
  Length - 30 inches.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Description. The rulers shall be as shown in Figure 1 and as specified herein. The edges of the ruler shall provide a smooth firm guide for a ruling pen and drafting pencil. When tested as specified in 4.5.2.1 and 4.5.2.2 the rulers shall be capable of being adjusted to form any smooth, irregular, or regular curve as desired and shall remain in a set position indefinitely without attaining a permanent set.

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3.2 First article.

3.2.1 First-produced ruler. The contractor shall furnish one or more rulers as specified (see 6.2), for examination and test within the time frame specified (see 6.2) to prove that his production methods will produce rulers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Corrosion-resisting metal. Exposed metals used in the construction of the rulers shall be corrosion-resisting steel, nickel-plated brass, or other suitable metals which are either corrosion resisting or suitably treated to resist corrosion, and shall show no signs of deterioration and corrosion or peeling or cracking of the protective coating when tested as specified in 4.5.2.1 and 4.5.2.3.

3.3.2 Rubber and plastic. When tested as specified in 4.5.2.1, natural rubber, synthetic rubber and plastic used as an encasing material or as a fabricating material shall show no signs of distortion, deterioration, or cracking.

3.4 Temperature conditions. When tested as specified in 4.5.2.1, the rulers shall not be damaged by storage at any ambient temperature from plus 145° F for a period of 4 hours daily to minus 50° F for periods of at least 3 days duration.

3.5 Identification marking. The rulers shall be identified in accordance with MIL-STD-130.

3.6 Finishing and painting. The equipment shall be finished and painted in accordance with commercial practice for this type of equipment.

3.7 Type I. Type I rulers shall be constructed with a soft metal core between two strips of flexible spring steel encased in an outer covering of natural or synthetic rubber. The ends of the ruler shall be protected with corrosion-resisting metal ferrules or clips.

3.8 Type II. Type II rulers shall consist of a soft metal bar encased in a corrosion-resisting metal coil. A flexible, corrosion-resisting metal ruling edge shall be attached to the coil by corrosion-resisting metal clips.

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3.9 Type III. Type III rulers shall be constructed from a perforated and slotted, rectangular strip of elastomer material which supports a flexible plastic ruling edge and maintains set curves by means of a soft metal rod inserted opposite the ruling edge.

3.10 Workmanship. All parts, components, and assemblies of the ruler, including molded parts and stampings shall be clean and free from scratches, nicks, burrs, rough edges, and other imperfections on the ruling edge.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced inspection.

4.3.1 Examination. The first-produced ruler shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced ruler shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II.

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4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 4.0 percent defective.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The rulers shall be examined as specified herein for the following defects:

101. Scratches, nicks, burrs, rough edges, and other imperfections on the ruling edge.
102. Identification marking missing, illegible, or not as specified.
103. Materials not as specified.

4.5.2 Tests.

4.5.2.1 Storage test. The ruler shall be subjected to a cycle of temperature and relative humidity extremes in the following order:

(a) A high temperature series consisting of three exposures of not less than 1 hour each on 3 successive days. Cycling shall start at ambient conditions, increase to not less than 145° F (and not less than 15 percent relative humidity), hold for a period of 1 hour, then return to ambient conditions.
(b) A low temperature condition consisting of not less than 72 hours exposure at a temperature of minus 50° F plus or minus 5° F.
(c) A high humidity condition consisting of not less than 24 hours exposure at a relative humidity of not less than 98 percent and a temperature of plus 80° plus or minus 2° F.
(d) A normal temperature and humidity condition consisting of not less than 8 hours at a temperature of not less than plus 65° F nor more than plus 85° F and a relative humidity of not less than 20 percent nor more than 80 percent.

4.5.2.1.1 Failure criteria. Nonconformance to 3.3.1, 3.3.2, and 3.4 shall constitute failure of this test.

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4.5.2.2 Adjustable radius test.

4.5.2.2.1 Type I and Type II. Type I and Type II rulers shall be adjusted to the circumference of an 8-inch diameter circle for not less than 180 degrees, examined for ability to maintain the adjustment, and then straightened and examined for permanent set. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.2.2 Type III. Type III rulers shall be tested as specified in 4.5.2.2.1, using a 5-inch diameter circle. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.3 Finish adherence test. Type II rulers shall be bent around a 4-inch diameter bar for 180 degrees plus or minus 2 degrees and then straightened rapidly. Repeat this procedure 10 times. Nonconformance to 3.1 and 3.3.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104.	Materials, methods, or containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
105.	Gross weight exceeds 200 pounds for Level A.
106.	Strapping not as specified for Level A.
107.	Gross weight or size exceeds the limitations of the box for Level B.
108.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Each ruler shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.2 Commercial. Rulers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The rulers of like description, preserved as specified in 5.1, shall be packed in quantities not to exceed 200 pounds in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. The box shall be lined with Weather-Resistant fiberboard. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Size 25 as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The rulers of like description, preserved as specified in 5.1, shall be packed in close-fitting fiberboard boxes conforming to PPP-B-636, V3c, style optional. The gross weight or size of each box shall not exceed the limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Rulers of like description, preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The rulers are intended for use in drawing smooth, regular, and irregular curved lines with a ruling pen, or drafting pencil.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and length of rulers required (see 1.2).

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(c) Time frame required for submission of first-produced ruler and number of rulers required (see 3.2.1).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

6.3 First-produced ruler. Any changes or deviations of production rulers from the approved first-produced ruler during production will be subject to the approval of the contracting officer. Approval of the first-produced ruler will not relieve the contractor of his obligation to furnish rulers conforming to this specification.

6.4 Information figures. Figure 1 shows types of adjustable curve rulers which have been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other rulers which conform to this specification.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

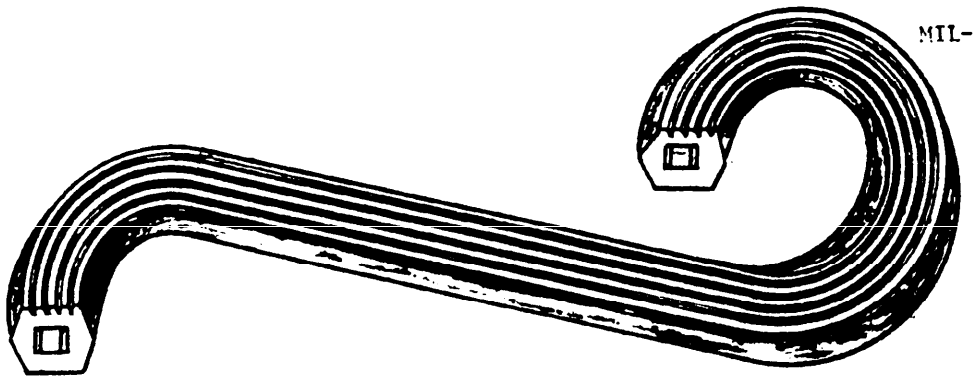
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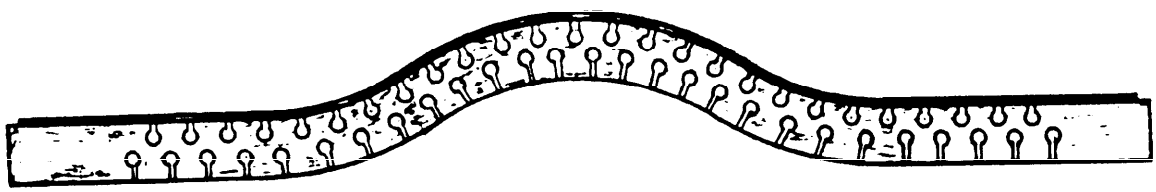
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Type I



Type II



Type III

Figure 1. Adjustable curve rulers

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-3471C Rulers, Drafting, Adjustable Curve

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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